

Transient Cooling Analysis: Water in Insulated Cylinder Enclosure

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1 Problem Statement

A volume of water initially at $T_0 = 90^\circ\text{C}$ is contained in a cylinder inside a plywood enclosure. We want to determine how the water temperature decreases over time as heat is lost to the ambient environment at $T_\infty = 20^\circ\text{C}$.

Table 1: System parameters

Parameter	Symbol	Value
Cylinder diameter	D	25 cm
Cylinder length	L	30 cm
Water volumes analyzed	V	5, 10, 15 liters
Initial temperature	T_0	90°C
Ambient temperature	T_∞	20°C
Water density	ρ	1000 kg/m^3
Water specific heat	c_p	$4186 \text{ J/(kg}\cdot\text{K)}$

2 Governing Equation

Using the lumped capacitance model (assuming uniform water temperature):

$$mc_p \frac{dT}{dt} = -\frac{T - T_\infty}{R_{total}} \quad (1)$$

where:

- m = water mass [kg]
- c_p = specific heat capacity [J/(kg·K)]
- R_{total} = total thermal resistance [K/W] (temperature-dependent due to radiation)

For constant R_{total} , the analytical solution is:

$$T(t) = T_\infty + (T_0 - T_\infty)e^{-t/\tau} \quad (2)$$

where the **time constant** is:

$$\tau = R_{total} \cdot m \cdot c_p = R_{total} \cdot C \quad (3)$$

3 Thermal Capacitance

Table 2: Thermal capacitance for different water volumes

Volume [L]	Mass [kg]	$C = mc_p$ [kJ/K]	Energy to cool [kJ]
5	5.0	20.9	1,465
10	10.0	41.9	2,930
15	15.0	62.8	4,395

The energy required to cool from 90°C to 20°C:

$$E = mc_p \Delta T = m \times 4186 \times 70 \text{ J} \quad (4)$$

4 Time Constant Analysis

From the steady-state analysis, the thermal resistance is approximately:

$$R_{total} \approx 0.75 - 0.95 \text{ K/W} \quad (5)$$

(varies with temperature due to radiation dependence)

Table 3: Time constants and cooling times

Volume [L]	τ [hours]	T at $t = \tau$ [°C]	Time to 60°C [hours]	Time to 40°C [hours]
5	5.5	45.8	2.6	6.3
10	10.0	45.8	5.2	12.6
15	14.5	45.8	7.9	18.9

Key insight: The time constant τ scales linearly with water mass. Doubling the water volume doubles the cooling time.

5 Temperature vs. Time

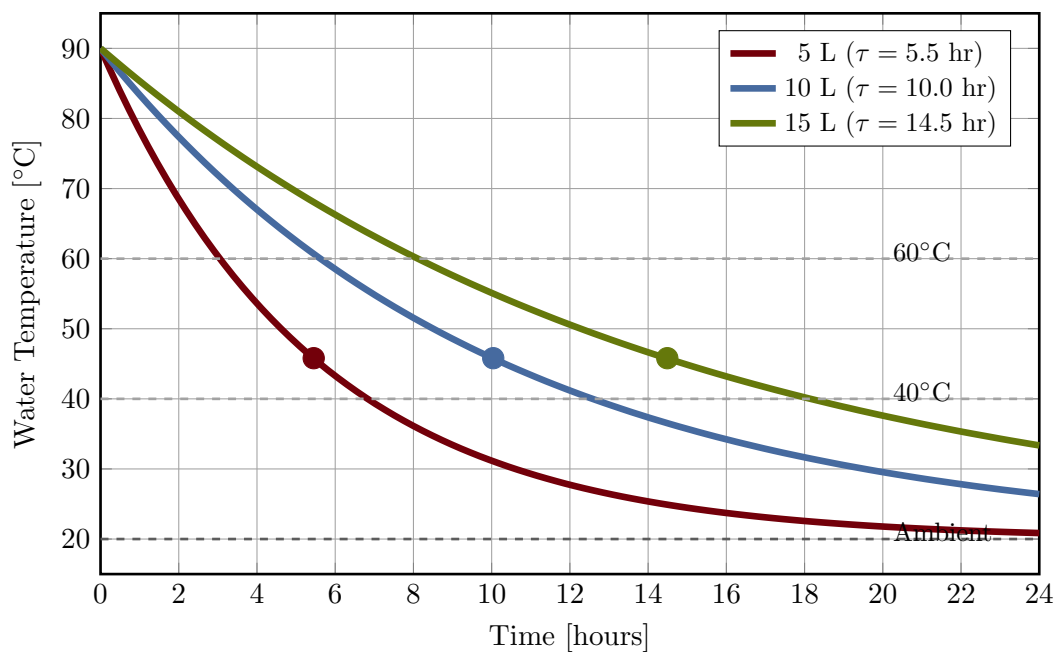


Figure 1: Water temperature vs. time for different volumes. Dots mark the time constant τ where $T = 45.8^\circ\text{C}$ (36.8% of ΔT remaining).

6 Heat Loss Rate vs. Time

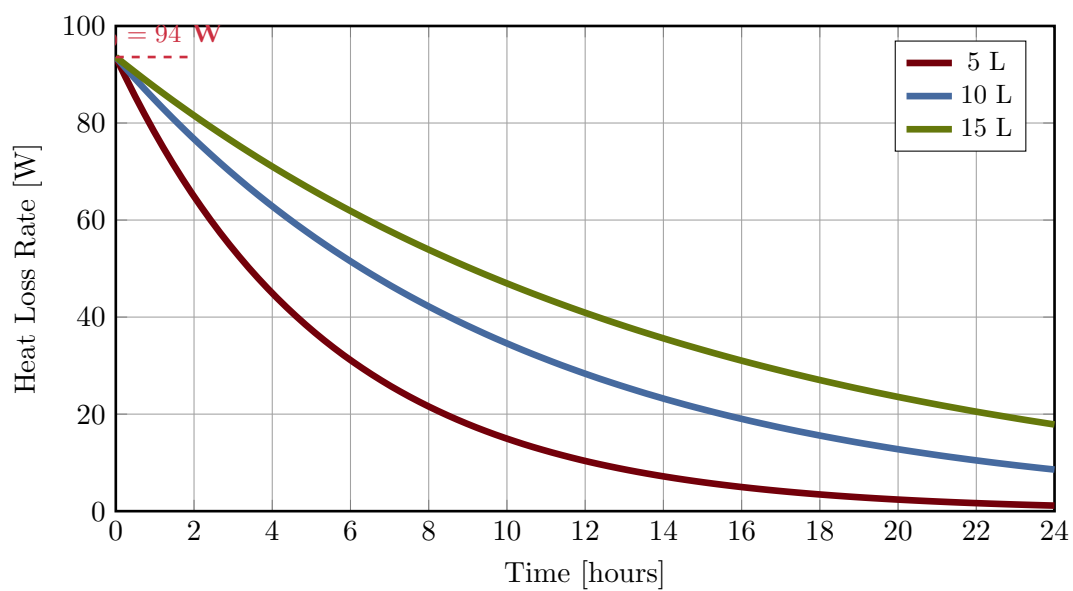


Figure 2: Heat loss rate decreases exponentially as the water cools. All curves start at $Q_0 \approx 94$ W but decay at different rates.

7 Cooling Timeline

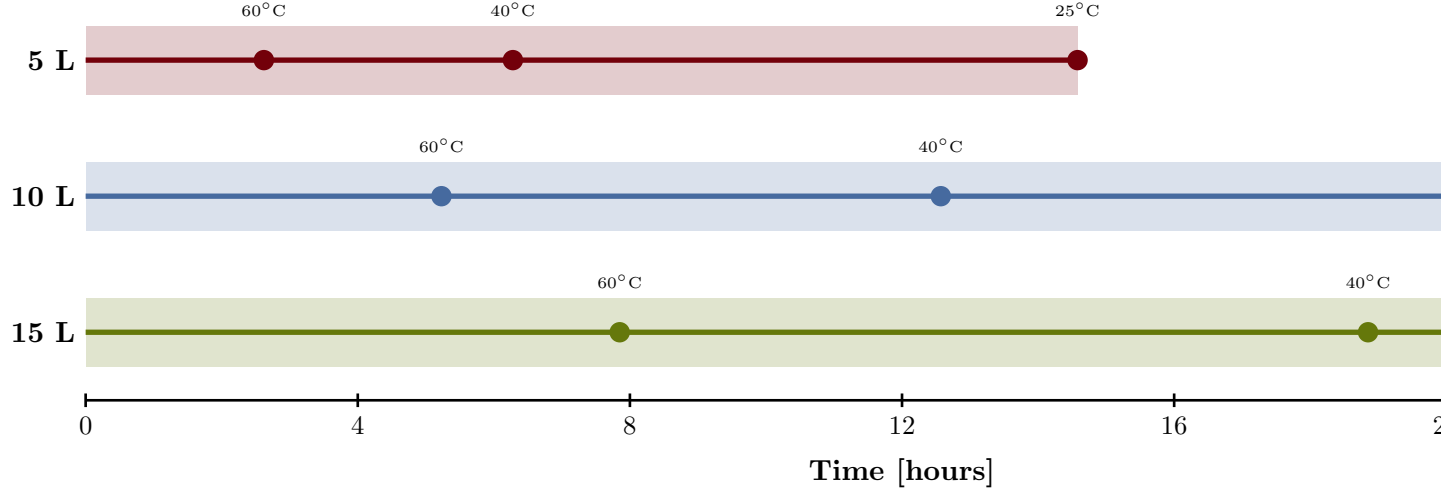


Figure 3: Cooling timeline showing when water reaches key temperatures.

8 Detailed Results

Table 4: Complete cooling timeline

Volume	Time to reach temperature			Temperature at time		
	60°C	40°C	25°C	2 hr	6 hr	12 hr
5 L	2.6 hr	6.3 hr	14.6 hr	69°C	44°C	29°C
10 L	5.2 hr	12.6 hr	29.2 hr	78°C	58°C	42°C
15 L	7.9 hr	18.9 hr	43.5 hr	81°C	65°C	50°C

9 Physical Interpretation

9.1 Why the Exponential Decay?

The cooling follows an exponential curve because:

$$\frac{dT}{dt} = -\frac{T - T_{\infty}}{\tau} \quad (6)$$

The rate of cooling is *proportional to the temperature difference*. When hot, the water loses heat quickly; as it approaches ambient, the heat loss slows dramatically.

9.2 Effect of Water Volume

- **Thermal capacitance** $C = mc_p$ increases linearly with mass
- **Heat loss rate** $Q = \Delta T/R$ is independent of mass (same at $t = 0$)
- **Time constant** $\tau = RC$ increases linearly with mass
- **Result:** More water = slower cooling, proportionally longer times

9.3 Practical Implications

For keeping water hot:

- 5 L of water drops below 60°C in **2.6 hours**
- 10 L stays above 60°C for **5.2 hours**
- 15 L stays above 60°C for **7.9 hours**

The plywood enclosure provides reasonable insulation—initial heat loss is only ~ 94 W compared to ~ 600 W that would occur with direct exposure to air (no enclosure).

10 Summary

Table 5: Key results summary

Metric	5 L	10 L	15 L
Water mass	5 kg	10 kg	15 kg
Thermal capacitance	20.9 kJ/K	41.9 kJ/K	62.8 kJ/K
Time constant τ	5.5 hr	10.0 hr	14.5 hr
Initial heat loss	94 W	94 W	94 W
Time to 60°C	2.6 hr	5.2 hr	7.9 hr
Time to 40°C	6.3 hr	12.6 hr	18.9 hr
Time to 25°C	14.6 hr	29.2 hr	43.5 hr

The time constant relationship is straightforward:

$$\tau \approx 1.1 \text{ hours per liter of water} \quad (7)$$